BSc in Network Security and Computer Forensics

Name of Programme	BSc in Network Security and Computer Forensics
Programme Overview	Interested in safeguarding digital systems and uncovering cybercrime? Our BSc in Network Security and Computer Forensics is a specialised degree designed to prepare students to protect information assets and investigate cyber incidents. The programme offers practical training in penetration testing, digital forensics, and network defence, combining fundamental computing with advanced security practices. Students will engage with key topics such as ethical hacking, cybercrime investigation, and malware analysis, while also developing a strong grounding in ethical and legal considerations. Graduates emerge ready for careers as network security specialists or digital forensic investigators in government, corporations, and law enforcement, addressing critical challenges in today's digital world.
Admission Requirements	IGCSE/equivalent with 5 Passes including English and Mathematics. Applicants in possession of a Diploma or Higher Diploma in related fields may be given exemptions based on the credit point equivalency.
Technical Requirements	To ensure a smooth and effective learning experience, students enrolling in this programme must meet the following technical requirements: Device Requirements Computer/Laptop: Windows (10 or later) or Mac (macOS 10.15 or later); Processor: Intel i3 (or equivalent) or above; RAM: Minimum 4 GB (8 GB recommended for multitasking); Storage: Minimum 20 GB free space for coursework and downloads; Camera and Microphone: Built-in or external for live sessions and presentations OR Tablet/Smartphone: Suitable for accessing content on the go (limited functionality for assignments/exams) Internet Requirements Connection speed: Minimum 5 Mbps download and 2 Mbps upload (10 Mbps recommended for seamless video conferencing); Stable Wi-Fi or Ethernet connection recommended for live Zoom sessions Software and Platforms Virtual Live Sessions: Zoom (synchronous learning); Document Processing: Microsoft Office (Word, Excel, PowerPoint) or equivalent (Google Docs) Browser Requirements Supported browsers: Chrome (latest version), Safari (for Mac users); Cookies and JavaScript: Enabled for full Blackboard functionality Additional Tools (Optional but Recommended) Headset: For clear audio during virtual classes; External storage (USB/Cloud): For backups of important coursework; Antivirus software: To protect against malware or data loss
Typical Full-Time Study Period	4 years
Number of Modules	37 (35 core + 2 elective)
Total Programmes Credits	480
Fees per Programme Credit	$13
Total Tuition Fee	$6240

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SEMESTER 1	
Computer and its Essentials 1	
Code: C5-CE1-20	This module provides a foundational understanding of computers and their essentials, covering basic functions, hardware and software components, and how they operate. It explores the history and evolution of computers and data processing. Students will learn about Microsoft Office, with an emphasis on Microsoft Word and PowerPoint. Through hands-on activities and assessments, learners will develop a comprehensive understanding of computer essentials.
Module Fees: $130	
Module Credits: 10	
Programming Logic and Design	
Code: C5-PLD-20	This module provides a foundation in programming principles, focusing on algorithm design, logical structures, and flowchart development. Students will learn problem-solving techniques essential for software development and cybersecurity applications. It covers key concepts such as variables, data types, control structures, and debugging, ensuring a solid understanding of programming fundamentals.
Module Fees: $130	
Module Credits: 10	
Mathematics	
Code: C5-MAT-20	This module covers essential mathematical concepts used in computing, including Boolean algebra, logic gates, number systems, and probability theory. Students will develop analytical skills necessary for network security, encryption, and algorithm design. The module also explores set theory, discrete mathematics, and binary arithmetic, providing a strong mathematical foundation for computing applications.
Module Fees: $130	
Module Credits: 10	
Operating Systems and Hardware	
Code: C5-OSH-11	Students will explore the fundamentals of operating systems, hardware configurations, and system management. The module covers key concepts such as process management, memory allocation, file systems, and security protocols. Through hands-on experience in managing system resources, virtualisation, and troubleshooting, students will develop practical skills essential for IT administration and cybersecurity.
Module Fees: $260	
Module Credits: 20	
Professional Issues and Ethics	
Code: C6-PIE-20	This module examines the ethical and legal issues in cybersecurity and computing, exploring topics such as data privacy, intellectual property rights, cybercrime laws, and professional responsibilities. Students will study cybersecurity policies, compliance regulations, and responsible digital practices, ensuring they develop ethical decision-making skills essential for technology and security professions.
Module Fees: $130	
Module Credits: 10	

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SEMESTER 2	
Networking Fundamentals	
Code: C6-NEF-20	This module provides an introduction to networking principles, covering network topology, protocols, IP addressing, and troubleshooting techniques. Students will explore key concepts such as LAN and WAN technologies, subnetting, routing, and network security fundamentals. This foundation prepares them for more advanced studies in networking, cybersecurity, and infrastructure management.
Prerequisite: C5-CE1-20	
Module Fees: $260	
Module Credits: 20	
Database Concepts	
Code: C6-DBC-20	This module explores the principles of database design, management, and security, covering data modelling, normalisation, and relational database concepts. Students will learn SQL for querying and managing data, as well as techniques for ensuring data integrity, access control, and protection against cyber threats, preparing them for database administration and security roles.
Module Fees: $130	
Module Credits: 10	
Academic Writing for STEM	
Code: D6-AWS-20	This specialised module enhances research and technical writing skills, focusing on academic documentation, citation techniques, and effective communication in STEM disciplines. Students will develop proficiency in structuring research papers, analysing scientific literature, and presenting technical information clearly and concisely, preparing them for academic and professional writing in technology fields.
Module Fees: $130	
Module Credits: 10	
Computer and its Essentials 2	
Code: C6-CE2-20	This module builds on foundational computing skills, equipping students with advanced capabilities in Microsoft Excel and Google collaboration tools. It covers key spreadsheet functionalities, including data organisation, advanced formulas, chart creation, and data analysis using pivot tables. Additionally, students will explore Google Docs, Spreadsheets, Forms, and Drive to enhance online collaboration and information sharing.
Prerequisite: C5-CE1-20	
Module Fees: $130	
Module Credits: 10	
Essentials of Linux	
Code: C6-EOL-20	This module introduces Linux operating systems, focusing on essential command-line operations, system security, and user management. Students will explore file system structure, permissions, and process management while gaining hands-on experience with open-source networking tools. The module also covers system administration tasks, preparing students for careers in Linux-based environments and security.
Prerequisite: C5-OSH-11	
Module Fees: $130	
Module Credits: 10	

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SEMESTER 3	
Computer Forensics	
Code: C6-CFO-23	Students will learn forensic investigation techniques, data recovery methods, and evidence collection strategies essential for cybercrime investigations. The module covers digital evidence handling, forensic imaging, malware analysis, and incident response procedures. Through practical exercises, students will develop the skills needed to analyse cyber incidents and support legal proceedings in cybersecurity.
Prerequisite: C5-OSH-11	
Co-requisite: C6-CFL-23	
Module Fees: $130	
Module Credits: 10	
Computer Forensics Lab	
Code: C6-CFL-23	This hands-on lab module provides practical experience in forensic investigation techniques, including disk imaging, data recovery, and malware analysis. Students will use industry-approved and standardised tools to examine digital evidence, detect security breaches, and analyse cyber threats. The module enhances critical thinking and technical skills essential for cybersecurity and forensic professionals.
Prerequisite: C5-OSH-11	
Co-requisite: C6-CFO-23	
Module Fees: $130	
Module Credits: 10	
Programming Using C++	
Code: C6-IPC-11	This module introduces students to C++ programming, covering programming logic, object-oriented principles, and software development methodologies for cybersecurity applications. Students will learn about data structures, memory management, and algorithm design while developing secure and efficient code. Hands-on projects reinforce problem-solving skills essential for software development and cybersecurity solutions.
Prerequisite: C5-PLD-20	
Module Fees: $260	
Module Credits: 20	
Routing and Switching	
Code: C6-RSW-20	This module covers network routing and switching principles, including VLANs, WAN, LAN, subnetting, and traffic management, essential for network security professionals. Students will learn to configure and troubleshoot network devices, implement security protocols, and optimise network performance. The module provides a strong foundation for designing, managing, and securing modern network infrastructures.
Prerequisite: C6-NEF-20	
Co-requisite: C6-RSL-20	
Module Fees: $130	
Module Credits: 10	
Routing and Switching Lab	
Code: C6-RSL-20	This practical networking module focuses on configuring, securing, and troubleshooting network routers and switches in simulated environments. Students will gain hands-on experience using industry-standard Cisco software and hardware, learning essential skills in network administration, traffic management, and security implementation. The module prepares students for real-world networking challenges in cybersecurity and IT infrastructure.
Prerequisite: C6-NEF-20	
Co-requisite: C6-RSW-20	
Module Fees: $130	
Module Credits: 10	

SEMESTER 4	
Scripting for Cyber Security	
Code: C6-SCS-25 Prerequisite: C6-IPC-11 Module Fees: $260 Module Credits: 20	Students will learn scripting techniques using Python, PowerShell, and Bash to automate security tasks such as threat detection, incident response, and vulnerability assessments. The module covers scripting fundamentals, regular expressions, and automation frameworks, equipping students with the skills to develop efficient security scripts and streamline cybersecurity operations.
Principles of Cyber Security	
Code: C6-PCS-25 Prerequisite: C6-NEF-20 Module Fees: $130 Module Credits: 10	An introduction to core cybersecurity principles, including risk assessment, threat modelling, and security frameworks used to protect digital assets. Students will explore attack vectors, defensive strategies, and ethical considerations in cybersecurity. This module builds foundational knowledge essential for understanding cybersecurity threats, vulnerabilities, and best practices in securing systems.
Operating System Forensics	
Code: C6-OSF-23 Prerequisite: C6-CFO-23 Module Fees: $130 Module Credits: 10	This module focuses on forensic techniques used to analyse and investigate security breaches within operating systems. Students will study file system analysis, memory forensics, log analysis, and malware detection. Practical exercises using forensic tools will enhance their ability to extract, preserve, and interpret digital evidence in cybercrime investigations.
Network Security	
Code: C6-NSE-20 Prerequisite: C6-NEF-20 Module Fees: $130 Module Credits: 10	Students will explore network defence strategies, including firewall configurations, intrusion detection systems, and encryption protocols. The module covers key topics such as VPNs, security policies, and secure network design. Practical labs ensure students develop hands-on skills in detecting and mitigating network-based cyber threats and vulnerabilities.
Wireless and Mobile Security	
Code: C6-WMS-23 Module Fees: $130 Module Credits: 10	This module examines the security risks in wireless networks, mobile devices, and IoT environments. Topics include wireless encryption, authentication protocols, access control mechanisms, and mobile application security. Students will gain hands-on experience in securing wireless infrastructures, detecting vulnerabilities, and implementing security policies for mobile and IoT devices.

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SEMESTER 5	
Information and Data Security	
Code: C7-IDS-23	Students will study data protection strategies, including access control models, cryptographic methods, and compliance with security standards. The module covers data classification, secure storage solutions, and identity management. Students will learn how to safeguard sensitive data against breaches, unauthorised access, and cyber threats in various computing environments.
Prerequisite: C6-CFO-23, C6-CFL-23	
Module Fees: $130	
Module Credits: 10	
Ethical Hacking	
Code: C7-EHK-13	A hands-on module covering penetration testing techniques, vulnerability assessments, and ethical hacking methodologies to identify and mitigate security threats. Students will explore network scanning, exploitation techniques, and countermeasures. The module emphasises ethical considerations, legal implications, and responsible disclosure in conducting cybersecurity assessments and penetration testing engagements.
Prerequisite: C6-PCS-23	
Module Fees: $260	
Module Credits: 20	
Software Defined Network Engineering	
Code: C7-SDN-20	This module introduces software-defined networking (SDN) concepts, exploring network automation, virtualisation, and programmable network infrastructure. Students will learn about SDN controllers, network function virtualisation (NFV), and security challenges in SDN environments. Hands-on labs provide experience in deploying, managing, and securing SDN architectures in modern networks.
Prerequisite: C6-RSW-20, C6-RSL-20	
Module Fees: $130	
Module Credits: 10	
Forensics Investigation Techniques	
Code: C7-FIT-23	Students will gain expertise in forensic methodologies, digital evidence handling, and cybercrime investigation frameworks. The module covers forensic imaging, log analysis, data recovery, and malware forensics. Practical exercises using industry-standard tools will enhance students' ability to conduct forensic investigations and support legal proceedings in cybersecurity incidents.
Prerequisite: C6-OSF-23	
Module Fees: $130	
Module Credits: 10	
Cryptographic Techniques	
Code: C7-CGT-23	This module covers encryption algorithms, key management, and secure communication techniques essential for cybersecurity professionals. Topics include symmetric and asymmetric encryption, digital signatures, and hashing techniques. Students will learn cryptographic protocols, secure data transmission methods, and cryptanalysis techniques used to protect and verify digital information.
Module Fees: $130	
Module Credits: 10	

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SEMESTER 6	
Research Methods for STEM	
Code: C7-RMS-20	An introduction to research methodologies, focusing on data analysis, hypothesis testing, and experimental design in computing and cybersecurity. Students will learn literature review techniques, research proposal development, and ethical considerations in research. This module prepares students for conducting independent studies and technical research in STEM fields.
Prerequisite: C5-MAT-20	
Module Fees: $130	
Module Credits: 10	
Malware Analysis	
Code: C7-MAN-25	Students will study malware behaviour, reverse engineering techniques, and threat intelligence analysis to combat cyber threats. The module covers static and dynamic malware analysis, sandboxing, and exploit detection. Practical labs provide experience in analysing real-world malware samples and developing strategies to mitigate and defend against cyberattacks.
Prerequisite: C6-NSE-20	
Module Fees: $260	
Module Credits: 20	
Advanced Ethical Hacking	
Code: C7-AEH-23	An advanced module that delves deeper into penetration testing, exploit development, and security bypass techniques. Students will explore buffer overflow attacks, privilege escalation, advanced reconnaissance, and post-exploitation tactics. Ethical and legal considerations are emphasised, ensuring students develop responsible hacking skills for cybersecurity defence.
Prerequisite: C7-EHK-13	
Module Fees: $130	
Module Credits: 10	
Mobile Forensics	
Code: C7-MFO-20	This module focuses on mobile device security, data extraction techniques, and forensic investigation of smartphones and tablets. Students will analyse mobile operating systems, encrypted communications, and deleted data recovery. Hands-on labs provide experience in using forensic tools to investigate mobile security incidents and cybercrime cases.
Prerequisite: C6-CFO-23	
Module Fees: $130	
Module Credits: 10	

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SEMESTER 6 ELECTIVE GROUP (CHOOSE ONE)	
Internet of Things	
Code: C7-IOT-23	This module explores IoT security challenges, device vulnerabilities, and secure implementation strategies in interconnected environments. Students will study IoT architectures, network protocols, and attack surfaces in smart devices. Hands-on labs will provide experience in securing IoT ecosystems, detecting threats, and implementing best practices to mitigate cyber risks in IoT networks.
Prerequisite: C6-NEF-20	
Module Fees: $130	
Module Credits: 10	
Cloud Computing and Security	
Code: C7-CCS-20	This module explores the principles of cloud computing and its associated security challenges. Students will learn about cloud architectures, virtualisation, data protection strategies, and compliance with cloud security frameworks. The course also covers risk management, identity and access control, and best practices for securing cloud environments from cyber threats.
Prerequisite: C6-NEF-20	
Module Fees: $130	
Module Credits: 10	
Security by Design	
Code: C7-SDE-20	This module focuses on integrating security measures at the core of software and system development. Students will study secure coding principles, threat modelling, and vulnerability mitigation techniques. The module emphasises proactive security strategies, ensuring that applications and networks are resilient against cyber threats from the early stages of development.
Prerequisite: C6-NSE-20	
Module Fees: $130	
Module Credits: 10	
SEMESTER 7	
Cyber Crime Investigation	
Code: C7-CCI-25	Students will explore legal and investigative approaches to cybercrime, including fraud, identity theft, and digital evidence collection. The module covers case analysis, cyber laws, forensic reporting, and courtroom procedures. Practical exercises will enhance students' ability to conduct cybercrime investigations and support law enforcement efforts.
Prerequisite: C7-FIT-23	
Module Fees: $130	
Module Credits: 10	
Research Project 1: Proposal Writing	
Code: C7-RP1-20	A preparatory module for research projects, where students develop research proposals and methodologies for cybersecurity topics. Students will learn academic writing, literature review techniques, and research design principles. This module ensures they are equipped to plan and justify their research effectively before conducting their studies.
Prerequisite: C7-RMS-20	
Module Fees: $130	
Module Credits: 10	
Cyber Law	
Code: C7-CYL-23	This module examines legal frameworks, data protection laws, and regulatory compliance in cybersecurity. Topics include intellectual property rights, cybercrime legislation, and digital privacy laws. Students will analyse case studies and learn how cybersecurity policies align with national and international legal standards.
Prerequisite: C6-PCS-25	
Module Fees: $130	
Module Credits: 10	
Entrepreneurship and Innovation	
Code: B8-ENI-20	The purpose of this module is to equip students with the knowledge and skills necessary to understand entrepreneurial processes and behaviours. It explores the fundamentals of entrepreneurship, including the creation and development of business ideas, and the essential steps in crafting a comprehensive business plan for successful ventures.
Module Fees: $260	
Module Credits: 20	

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SEMESTER 7 ELECTIVE GROUP (CHOOSE ONE)	
Media and Storage	
Code: C8-MAS-20	This module covers digital media storage technologies, data retention policies, and storage security. Students will explore topics such as file systems, disk encryption, cloud storage solutions, and secure data backup methodologies. Practical exercises will help students understand how to manage, protect, and recover digital assets efficiently and securely.
Prerequisite: C6-CFO-23	
Module Fees: $130	
Module Credits: 10	
Media Forensics	
Code: C8-MEF-23	This module focuses on the investigation and analysis of digital media, including images, videos, and audio files. Students will learn forensic techniques for detecting tampering, recovering deleted media, and verifying the authenticity of digital evidence. The module prepares students for forensic analysis roles in cybercrime investigations.
Prerequisite: C6-CFO-23	
Module Fees: $130	
Module Credits: 10	
Information Security Management and Governance	
Code: C8-ISG-23	This module provides a comprehensive understanding of information security policies, risk management, and compliance with cybersecurity regulations. Students will explore governance frameworks such as ISO 27001 and NIST, learning how to develop and implement security policies that align with organisational and legal requirements.
Prerequisite: C6-PIE-20	
Module Fees: $130	
Module Credits: 10	
SEMESTER 8	
Research Project 2: Dissertation	
Code: C8-RP2-20	A capstone project requiring students to conduct independent research, analyse cybersecurity challenges, and present findings. Students will apply research methodologies, collect data, and produce a well-structured dissertation. This module develops their analytical and problem-solving skills in addressing real-world cybersecurity issues.
Prerequisite: C7-RP1-20	
Module Fees: $260	
Module Credits: 20	
Professional Practice in Computing	
Code: C7-PPC-20	A work-integrated learning module that provides students with industry exposure, project management skills, and hands-on cybersecurity experience. Students will work on real-world projects, collaborate with industry professionals, and develop technical and professional competencies essential for their careers in cybersecurity and computing.
Prerequisite: 240 credits	
Module Fees: $520	
Module Credits: 40	